VO XUAN NHI TRAN

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ABOUT ME

Third-year Software Engineering student at the University of Sydney with hands-on experience building and deploying full-stack applications and AI-powered tools. Skilled in Python, TypeScript, React, FastAPI, PostgreSQL and AWS, with practical experience in automated testing, debugging and cloud deployment. Interested in building reliable software that solves real-world problems.

EDUCATION

University of Sydney July 2024 – Present | Sydney, Australia

Bachelor of Engineering Honours (Software Engineering) - Dalyell Scholar - Vice Chancellor's International Scholarship

- **WAM:** 77.75/100 (Distinction Average) | **Expected Graduation:** 2028
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Agile Software Development, Security Engineering, Software Construction and Design, Data and Information Management

PROJECTS

RAG Document Assistant GitHub | Live Demo

Technologies: Python, FastAPI, React, PostgreSQL/pgvector, Docker, AWS

- Built a full-stack RAG app (React, FastAPI, PostgreSQL/pgvector, Gemini API) where users upload PDFs and get answers grounded in their documents; deployed on AWS EC2 with Nginx and CI/CD, debugging CORS and HTTPS mixed-content issues along the way.
- Wrote 13 pytest tests with mocked LLM calls and an isolated test database, including cross-session isolation tests that verify one user's documents are never retrievable by another.

JobCompass (Futura Remix Hackathon 2026) GitHub | Live Demo

Technologies: Next.js, TypeScript, Gemini API, Upstash Redis, PostHog, Tailwind CSS, Vitest

- Built an AI visa-eligibility screener for international students that outputs explainable APPLY / TAILOR / SKIP decisions backed by exact quotes from the job ad.
- Paired a deterministic rules engine with Gemini extraction, dropping any claim without direct evidence to prevent hallucination.
- Raised extraction accuracy from 60% on the hold-out set to 66.67% by rewriting the logic around clause structure (50 job ads: 35 dev, 15 hold-out).
- Tracked live analytics via PostHog, monitoring 32 WAU and optimizing analysis performance to maintain sub-1.3m latency.

RenoPilot – Roof Replacement App | Capstone Project with Real Industry Client in team of 8

Technologies: Next.js, React, TypeScript, Payload CMS, PostgreSQL

- Built the four-option landing page by extending the shared AppLanding component without breaking the other RenoPilot apps, then added persistent light/dark themes across the quote workflow (merged, client-accepted).
- Regression-tested the Property → Map → Materials → Quote flow and Undo feature (148 tests passing), reporting bugs to feature owners until fixed.
- Rotated through Tester Lead, Programmer Lead and Customer Liaison, presenting progress to the client.

YouTube Video Dubber (Chinese to Vietnamese) GitHub

Technologies: Python, faster-whisper, Gemini API, ffmpeg

- Automated a Chinese-to-Vietnamese dubbing pipeline (faster-whisper transcribe, Gemini translate, TTS, ffmpeg merge).
- Fixed a 35% audio overrun with a word-budget translation step based on measured TTS timings and silence trimming, reaching 0.00s drift across 13 blocks on a 72-second clip.
- Added caching and retry/backoff to work around Gemini free-tier quota limits.

TECHNICAL SKILLS

- **Languages:** Python, Java, TypeScript/JavaScript, SQL, HTML/CSS
- **Frameworks:** React, Next.js, FastAPI, Django, Django REST Framework
- **Databases:** PostgreSQL, pgvector, SQLite
- **Cloud & DevOps:** Docker, AWS EC2, Nginx, Vercel, Render, Linux
- **Tools:** Git/GitHub, pytest, JUnit, Gemini API